

The 25-Year Evolution of Lithium as a Disease-Modifying Agent in Dementia

A Narrative Review

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IMPORTANCE Lithium, a long-established cornerstone therapy for bipolar disorder, is a biologically plausible disease-modifying agent for neurodegenerative disorders, including mild cognitive impairment (MCI) and Alzheimer disease (AD).

OBSERVATIONS Rather than targeting a single pathology like amyloid or tau, lithium acts across multiple cellular resilience pathways. Chronic lithium exposure induces the anti-apoptotic protein B-cell lymphoma 2 (Bcl-2), enhances brain-derived neurotrophic factor (BDNF) signaling, inhibits glycogen synthase kinase-3 β (GSK-3 β), stabilizes mitochondrial function, and reduces oxidative stress. These convergent mechanisms promote neuronal survival and synaptic integrity. In humans, proton magnetic resonance spectroscopy studies found that lithium increased *N*-acetylaspartate levels, consistent with improved neuronal viability, and structural magnetic resonance imaging (MRI) studies found that lithium preserved gray matter and/or reversed illness-related atrophy in hippocampal and corticolimbic regions. In addition, extensive evidence demonstrates that low-dose lithium (approximately 0.3mM)—significantly lower than traditional psychiatric doses (0.6-1.0mM)—exerts robust neurotrophic and neuroprotective effects. Preclinical models have found that these concentrations stimulate hippocampal neurogenesis, promote structural plasticity, and protect against proteotoxic injury. Furthermore, epidemiological studies have associated cumulative lithium exposure with reduced dementia risk, and early randomized clinical trials in MCI suggest cognitive stabilization and favorable tau biomarker changes at low, well-tolerated doses. The recent repletion hypothesis suggests that lithium may also function as a physiological trace element, but these findings await independent replication.

CONCLUSIONS AND RELEVANCE These convergent data support a prospective clinical trial of low-dose lithium orotate to slow disease progression in MCI. Such an approach would prioritize established neuroprotective mechanisms while potentially mitigating the kidney and thyroid risks associated with higher-dose carbonate formulations. If low-dose lithium can indeed meaningfully alter disease trajectory, it would represent a much-needed, accessible, and inexpensive treatment that may be especially relevant in low- and middle-income countries.

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Lithium occupies a unique and historically important position in neuropsychiatry. As the first effective pharmacological treatment for bipolar disorder (BD), lithium helped establish the biological basis of psychiatric therapeutics. For decades, however, it was viewed narrowly as a mood stabilizer.¹ Over time, and especially recently, this perspective has broadened. Cellular, neurobiological, and translational studies now indicate that lithium's efficacy derives not from isolated neurotransmitter effects but from modulating upstream signaling networks that regulate apoptosis, mitochondrial bioenergetics, oxidative stress, and neurotrophic support.²⁻⁴

Importantly, these mechanisms intersect directly with the biology of key neurodegenerative diseases. Alzheimer disease (AD),

for instance, is increasingly understood as a disease not solely associated with amyloid or tau accumulation but as a progressive cellular vulnerability marked by mitochondrial dysfunction, impaired trophic signaling, synaptic loss, and apoptotic activation. Lithium's targets—B-cell lymphoma 2 (Bcl-2), brain-derived neurotrophic factor (BDNF), and glycogen synthase kinase-3 (GSK-3)—sit at central nodes within these pathways.⁵⁻⁷

Thus, 75 years after its therapeutic emergence, lithium may again be poised to exert a potentially powerful public health impact. In this narrative review, we synthesize evidence spanning cellular biology, experimental neuroprotection, human neuroimaging, epidemiology, and early clinical trials to evaluate lithium's evolution from mood stabilizer to candidate neuroprotective intervention; the inter-

ested reader is referred to several outstanding reviews and the references therein.^{2,3,8-13}

Neuroprogressive Models of Bipolar Disorder

Although this narrative review focuses on dementia, neuroprogressive models of BD provide a useful framework for understanding lithium's broader neurotrophic effects. Increasing evidence suggests that BD is not purely episodic but involves progressive neurobiological changes, including mitochondrial dysfunction, impaired neurotrophic signaling, and structural brain alterations. In this model—often termed *neuroprogression*—recurrent mood episodes contribute to cumulative neurobiological injury and functional decline.^{14,15} Notably, lithium appears distinctive among mood stabilizers in its ability to counter these processes, largely via its ability to preserve—and in some cases reverse—illness-related hippocampal and prefrontal cortical atrophy.¹⁶⁻²⁰ The neuroprogression framework also highlights the importance of timing. Lithium appears most effective when initiated early in the course of BD, when neurobiological damage is less advanced.

Mechanistic Foundations

Neurotrophic and Anti-Apoptotic Signaling

The demonstration that chronic lithium robustly upregulates the anti-apoptotic protein Bcl-2 in the mammalian brain⁴ became a conceptual inflection point for psychopharmacology. By linking a widely used mood stabilizer to durable activation of intrinsic neuronal survival pathways, this work reframed lithium as a neurotrophic and neuroprotective agent rather than solely a symptomatic treatment. While not all studies have replicated the observation of lithium-induced Bcl-2 upregulation (see Damiano and colleagues²¹ for a thoughtful discussion about methodological considerations), the preponderance of the evidence has confirmed that chronic lithium treatment is associated with increased Bcl-2 levels.^{5,22-27} This finding catalyzed research implicating cellular resilience in the pathophysiology of mood disorders, shifted expectations toward lithium's ability for structural and functional disease modification, and established a biological framework via which psychiatric therapeutics could be evaluated by their capacity to restore neuronal viability and long-term brain health.

Importantly, Bcl-2 upregulation does not occur in isolation but rather appears to be embedded within a broader neurotrophic signaling program. Lithium increases the transcription of BDNF, a master regulator of neuronal survival, synaptic plasticity, and circuit remodeling.^{6,28,29} Chronic lithium treatment elevates BDNF messenger RNA and protein levels in the hippocampus and frontal cortex^{6,30} in a manner consistent with activity-dependent neuroplasticity mechanisms.²⁸

Through tropomyosin receptor kinase B (TrkB) receptor signaling, BDNF activates phosphoinositide 3-kinase (PI3K)–protein kinase B (Akt) and extracellular signal-regulated kinase (ERK) cascades that directly suppress intrinsic apoptotic machinery while further increasing Bcl-2 expression.⁶ In this way, BDNF and Bcl-2 form a reinforcing trophic loop: BDNF promotes Bcl-2-mediated mitochondrial stability, while Bcl-2 enhances the survival environment that sustains neurotrophic signaling.

GSK-3 β Inhibition

GSK-3 β is constitutively active under baseline conditions and promotes pro-apoptotic and antiplasticity signaling. Seminal studies^{7,31} identified GSK-3 as a novel target for lithium's actions, with lithium inhibiting GSK-3 β through 2 complementary mechanisms. It directly competes with magnesium at the enzyme's catalytic site, reducing its activity at therapeutically relevant concentrations.³¹ Lithium also indirectly increases inhibitory phosphorylation of GSK-3 β .^{11,32} Together, these mechanisms result in sustained suppression of GSK-3 β activity. GSK-3 inhibition stabilizes β -catenin, enhances cyclic adenosine monophosphate response element-binding protein (CREB)–dependent transcription, and promotes expression of survival genes.¹¹ In animal models, GSK-3 β inhibition by lithium reduces tau hyperphosphorylation, attenuates amyloid- β production, regulates microglial activation and inflammatory signaling, and prevents neurodegeneration.^{8,33,34}

Mitochondrial Preservation and Bioenergetics

The upstream cellular changes just outlined converge most directly on the mitochondrion, the central regulator of neuronal energy metabolism, calcium buffering, oxidative stress, and apoptosis; this results in protection against diverse insults (see below). It should be noted that mitochondria—while best known for regulating neuronal survival and resilience—also play a critical role in regulating synaptic function (see Manji and colleagues³⁵ for expanded discussion of these topics).

Neuroprotective Effects Across Experimental Systems: Convergent Evidence From In Vitro and In Vivo Models

While the induction of Bcl-2, the activation of BDNF signaling, and the inhibition of GSK-3 provide a coherent intracellular framework for lithium's actions, the biological importance of this cascade is most clearly demonstrated by the breadth of its neuroprotective effects across experimental paradigms.

In Vitro Protection Against Excitotoxic and Apoptotic Stress

Chronic, but not acute, lithium exposure robustly protects cultured cerebellar granule neurons from glutamate- and NMDA-induced cell death, reducing calcium influx and preventing apoptotic cascades.³⁶ Protection correlates with marked increases in Bcl-2 expression and reductions in p53 and Bcl-2-associated X protein (BAX), directly linking lithium's anti-apoptotic gene regulation to functional survival.²³

Lithium also attenuates toxicity mediated by intracellular calcium dysregulation, a central feature of both BD pathophysiology and excitotoxic injury. In some models, lithium increased expression of Bcl-2 and endoplasmic reticulum (ER) chaperone proteins like GRP78/94, stabilized calcium handling, and prevented apoptosis.²⁴

Protection in Models of Ischemia and Metabolic Stress

In rodent models of focal cerebral ischemia, lithium treatment significantly reduced infarct volume, preserved neuronal architecture, and improved behavioral outcomes.^{8,22} These effects were accompanied by increased Bcl-2 expression, reduced cytochrome C release, decreased caspase-3 activation, and enhanced Akt signaling, again connecting gene regulation to tissue preservation.

In models of oxygen-glucose deprivation, lithium reduced neuronal loss and maintained electrophysiological function; lithium

increases intracellular glutathione levels, upregulates antioxidant enzymes, and suppresses lipid peroxidation and protein oxidation markers,^{37,38} suggesting that it also enhances intrinsic antioxidant capacity.

Protection Against Neurodegenerative and Proteotoxic Insults

Lithium has also been found to confer substantial protection in models of chronic neurodegeneration. For instance, lithium attenuated β -amyloid-induced toxicity, reduced tau phosphorylation, limited inflammatory cascades, and improved neuronal survival in models relevant to AD.^{8,10,33} It has also shown similar benefits in models of Huntington disease, amyotrophic lateral sclerosis, and spinal cord injury, where it promoted neuronal survival and enhanced regenerative responses.^{27,39}

What is striking across these experimental systems is the consistency of protection despite fundamentally different initiating insults—that is, lithium repeatedly enhanced survival regardless of whether neurons were challenged by excessive glutamate, calcium overload, oxidative stress, mitochondrial toxins, ischemia, or misfolded proteins. Thus, lithium appears to modulate upstream decision points that determine whether neurons adapt or undergo apoptosis when stressed.^{2,10}

Human Neuroimaging Evidence: Metabolic and Structural Correlates of Lithium's Neurotrophic Effects

Although preclinical studies have long shown lithium's cytoprotective and neurotrophic actions, the key translational question is whether these effects are detectable in the living human brain. Convergent proton magnetic resonance spectroscopy (¹H-MRS) and structural MRI studies have associated chronic lithium exposure with higher neuronal viability markers and greater corticolimbic gray matter volume.

MRS: Increases in Neuronal Viability (*N*-Acetylaspartate)

The earliest in vivo evidence came from ¹H-MRS studies quantifying *N*-acetylaspartate (NAA). Because NAA depends on oxidative phosphorylation and mitochondrial integrity, it is regarded as a marker of neuronal viability and bioenergetic function.⁴⁰ NAA is reduced in disorders marked by metabolic stress or neuronal compromise, including stroke, multiple sclerosis, amyotrophic lateral sclerosis, mitochondrial encephalopathies, and HIV-associated dementia; increases therefore generally indicate improved neuronal health and function (see articles cited here^{12,40–42}).

In a seminal longitudinal ¹H-MRS study of medication-free individuals with BD, Moore and colleagues⁴² found that 4 weeks of lithium increased frontal cortical NAA levels, providing the first direct in vivo evidence that lithium enhances neuronal viability in humans. The time course paralleled lithium's delayed clinical response, arguing against an acute osmotic effect. Subsequent studies extended these findings. For instance, one study reported higher NAA levels in lithium-treated vs unmedicated patients with BD,⁴³ suggesting persistence during maintenance therapy, and another study found NAA increases or normalization in the anterior cingulate and prefrontal cortex (PFC).⁴¹ In several cohorts, NAA changes correlated with clinical improvement, supporting metabolic recovery as part of therapeutic response.^{12,41,42}

Because NAA synthesis occurs in neuronal mitochondria and falls in line with impaired respiratory chain activity, increased NAA pro-

vides an indirect but biologically coherent marker of enhanced mitochondrial function. Together, these neuroimaging data offer a translational biomarker linking lithium's intracellular survival signaling to measurable metabolic improvement in vivo.

Structural MRI: Increases in Gray Matter Volume

If lithium enhances neuronal survival and trophic signaling, one would expect structural preservation or reversal of illness-related atrophy; multiple volumetric MRI studies support this, showing increased total and regional gray matter volumes. Early cross-sectional work showed marked (approximately 30%-40%) reductions in subgenual PFC volume in individuals with a family history of mood disorders; notably, patients receiving chronic lithium did not show these deficits, raising the possibility that lithium could mitigate illness-related atrophy.^{44,45} Consistent with lithium's broader cytoprotective effects, postmortem follow-up found glial reductions in the subgenual PFC.^{44,46}

More definitive evidence has come from longitudinal within-participant studies. For instance, increased total gray matter volume, particularly in the PFC, was found in medication-free patients with BD after about 4 weeks of lithium monotherapy.¹³ White matter and tissue-water measures were unchanged, arguing against edema or nonspecific swelling and suggesting neurotrophic effects. Voxel-based morphometry studies subsequently showed cortical gray matter increases after chronic lithium administration, especially in anterior cingulate, orbitofrontal, and medial temporal regions.^{12,47}

Integrating Imaging With Molecular Mechanisms

Taken together, the MRS and volumetric MRI data align well with lithium's cellular effects. These measurable structural and metabolic signatures are consistent with chronic lithium's neurotrophic effects in the living human brain, enhancing resilience of vulnerable circuits over time. The Box and Figure 1 summarize the extensive convergent preclinical and clinical data that highlight lithium's neurotrophic and neuroprotective effects.

Neurotrophic Effects Seen With Low-Dose Lithium

The evidence reviewed above is compelling; however, a potential concern is that the lithium doses generally used to treat BD (typically 0.6-1.0mM) are associated with a high adverse effect burden, and older individuals may be more sensitive to these.¹ It is thus noteworthy that several preclinical studies demonstrated neurotrophic and neuroprotective effects at levels substantially lower than those traditionally used to treat BD. Low-dose lithium (approximately 0.3mM) also regulates neurotrophic signaling, protects against neurodegeneration, and promotes structural plasticity, both in vitro and in vivo in the rodent brain (see the Box and the articles cited here^{48–53} and references therein).

Human Evidence Linking Lithium to Dementia Risk

The neurotrophic and cytoprotective properties of lithium previously described naturally raise the question of whether these same mechanisms could translate into protection against neurodegeneration in humans. While the pathophysiologies of BD and AD are clearly distinct, lithium exerts major effects on the most fundamental, critical, and evolutionarily conserved pathways; these fundamental neurotrophic mechanisms would be expected to be rel-

Box. Preclinical and Clinical Evidence for Lithium's Neurotrophic or Neuroprotective Effects**Preclinical Evidence**

- Preclinical models (rodent and *Drosophila*) have established that lithium's initial primary target is the potent inhibition of GSK-3 β .
- Preclinical models have established that, at therapeutically relevant concentrations, chronic lithium robustly increases levels of the major neuroprotective protein Bcl-2.
- Amyloid and tau: by inhibiting GSK-3 β , lithium reduces the production of amyloid β and prevents the hyperphosphorylation of tau protein (the precursor to neurofibrillary tangles).
- Neurogenesis: lithium stimulates the Wnt/ β -catenin pathway and increases levels of BDNF and Bcl-2, leading to a 25% increase in new hippocampal neurons in rodent models.
- Mitochondrial stability: chronic lithium upregulates Bcl-2 (anti-apoptotic) and preserves mitochondrial calcium homeostasis, protecting cells from glutamate excitotoxicity.
- Mitochondrial function: chronic lithium upregulates Bcl-2-associated athanogene, which is a chaperone protein that regulates glucocorticoid receptor translocation to nucleus and mitochondria. These effects preserve mitochondrial calcium homeostasis, protecting cells from excessive glucocorticoid exposure (ie, as seen in chronic stress) and related excitotoxicity and neuronal vulnerability.

Selected Preclinical Evidence of Low-Dose Lithium (Approximately 0.3-0.5mM)

- Trophic signaling: at concentrations of approximately 0.3mM, lithium enhances BDNF signaling and TrkB receptor expression, promoting neuronal survival and synaptic plasticity.
- Neurogenesis and structure: subtherapeutic doses stimulate the Wnt/ β -catenin pathway, significantly increasing hippocampal neurogenesis and dendritic branching complexity.
- Tau kinase modulation: low-dose treatment effectively decreases tau kinase activity and reduces neuropathological markers in senescence-accelerated and APP mutant mouse models.
- Targeted protection: low-dose lithium prevents the association of the p75NTR-sorilin receptor complex, attenuating proNGF-induced neuronal death.

Clinical Evidence: Structural and Functional Restoration

- In humans, lithium has demonstrated a unique ability to physically alter brain structure.
- Gray matter increases or normalization: landmark MRI studies showed an approximately 3% increase in total gray matter volume after only 4 weeks of treatment, peaking at 12 weeks.
- Gray matter increases or normalization: landmark MRI studies found that lithium-induced gray matter increases were seen in patients, not healthy volunteers, suggesting a lithium-induced reversal of illness-related atrophy.
- Gray matter preservation: MRI studies found that patients with bipolar disorder not treated with chronic lithium showed hippocampal atrophy; these effects were not seen in patients with bipolar disorder treated with chronic lithium.
- Metabolic recovery: magnetic resonance spectroscopy imaging revealed significant increases in *N*-acetylaspartate levels, indicating restored neuronal viability and mitochondrial health.
- Epidemiological correlation: large-scale population studies demonstrated that regions with higher trace levels of lithium in public drinking water have consistently lower rates of dementia and suicide.

Abbreviations: APP, amyloid precursor protein; Bcl-2, B-cell lymphoma 2; BDNF, brain-derived neurotrophic factor; GSK-3 β , glycogen synthase kinase-3 β ; MRI, magnetic resonance imaging; proNGF, pro-nerve growth factor; TrkB, tropomyosin receptor kinase B.

evant to AD, given that lithium's effects intersect directly with some core pathogenic mechanisms of AD. In parallel, lithium's mitochondrial stabilization and antioxidant effects would be expected to preserve neuronal bioenergetics.^{37,54}

Initial hints that lithium might protect against neurodegeneration emerged from naturalistic and registry-based studies of patients with BD. Several groups reported that individuals receiving long-term lithium therapy appeared to have lower-than-expected rates of dementia compared with both the general population and with patients with BD treated with other agents. A large Danish nationwide registry analysis⁵⁵ found that cumulative lithium exposure was associated with a significantly reduced incidence of dementia, whereas anticonvulsants and antipsychotics were not protective. This relationship showed a time-dependent pattern, with longer lithium exposure conferring greater reduction in dementia risk. Additional replication in Scandinavian and UK datasets lends consistency across health care systems and analytic approaches.⁵⁶ Ecological studies comparing regional lithium concentrations in drinking water found lower dementia prevalence in areas with trace environmental lithium exposure, suggesting that even low-level lithium exposure may exert neurobiological effects relevant to neurodegeneration.⁵⁵ However, these findings are inherently indirect and susceptible to confounding.

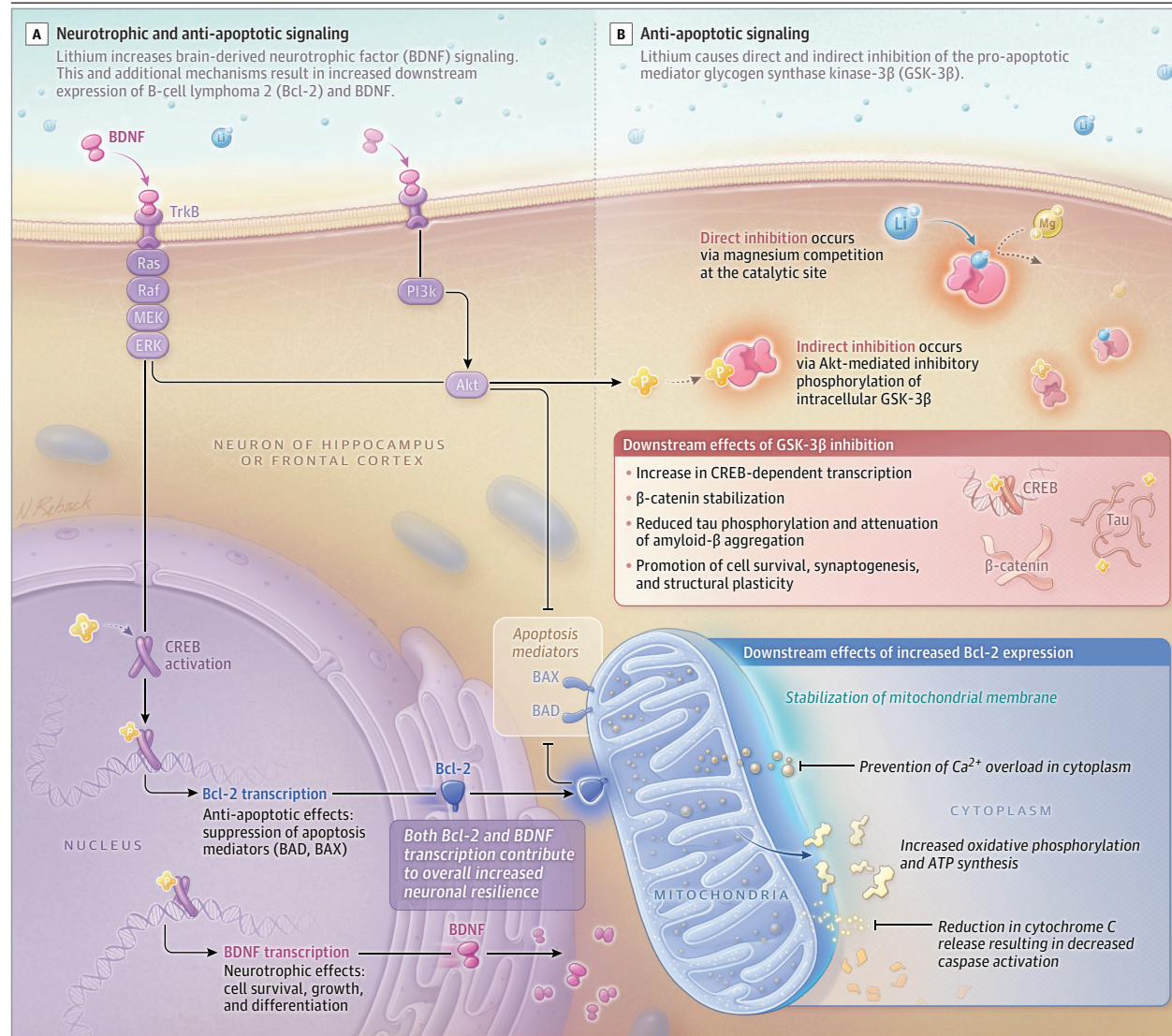
Evidence from randomized clinical trials has been more mixed. Much of the negative literature likely reflects studies of short treatment duration, advanced disease stage, or behavioral end points rather than neurodegeneration. These limitations constrain the ability to test lithium's hypothesized neuroprotective mechanisms, which are more likely to influence disease progression over months to years rather than weeks. Disease stage may also be critical. For instance, several studies that reported favorable signals focused on amnesic MCI or early AD with longer treatment periods (12-24 months), whereas many negative trials enrolled patients with established dementia.⁵⁷⁻⁶³

More recent studies have shifted toward exploring low-dose lithium. A small double-blind randomized clinical trial in patients with amnesic MCI reported that low-dose lithium (target serum levels approximately 0.25-0.5 mEq/L [to convert lithium from mEq/L to mmol/L, multiply by 1]) slowed cognitive decline over 12 months and reduced progression to dementia.⁶¹ Another longer-term trial suggested that low-dose lithium could maintain cognitive stability and influence cerebrospinal fluid biomarkers in patients with amnesic MCI over 2 to 3 years.⁵⁸ However, these studies were limited by small sample sizes and single-center designs. In addition, the LATTICE trial (2026), a 2-year pilot study, found that while lithium carbonate did not meet all its primary end points, it significantly slowed the decline of verbal memory (as measured by the California Verbal Learning Test Second Edition).⁵⁹

Lithium as a Trace Element in Brain Health: Implications for Dementia Prevention

A 2025 study advanced the provocative hypothesis that lithium may function as a physiological trace element required for normal brain health and, furthermore, that localized lithium deficiency may contribute directly to the pathophysiology of AD.^{64,65} Using high-sensitivity inductively coupled plasma mass spectrometry, the investigators performed an unbiased survey of metals in postmortem brain tissue from individuals with normal cognition, MCI, and

Figure 1. Neurotrophic and Neuroprotective Signaling Pathways Engaged by Lithium



A, Lithium engages convergent intracellular signaling cascades that promote neuronal survival, synaptic plasticity, and mitochondrial resilience. At the cell surface, lithium enhances brain-derived neurotrophic factor (BDNF)-tropomyosin receptor kinase B (TrkB) signaling, activating the extracellular signal-regulated kinase/mitogen-activated protein kinase (ERK/MAPIK) and phosphoinositide 3-kinase (PI3K)-protein kinase B (Akt) pathways. These cascades stimulate cyclic adenosine monophosphate response element-binding protein (CREB)-dependent transcription of neurotrophic and anti-apoptotic genes, including *Bcl-2* and *BDNF*, while suppressing pro-apoptotic mediators, such as B-cell lymphoma 2 (Bcl-2)-associated death promoter (BAD) and Bcl-2-associated X protein (BAX). B, Lithium directly and indirectly inhibits glycogen synthase kinase 3 (GSK-3β) through magnesium competition at the catalytic site and Akt-mediated inhibitory phosphorylation. GSK-3β inhibition stabilizes β-catenin, enhances CREB activity, and promotes transcriptional programs supporting synaptogenesis, cellular survival, and structural plasticity.

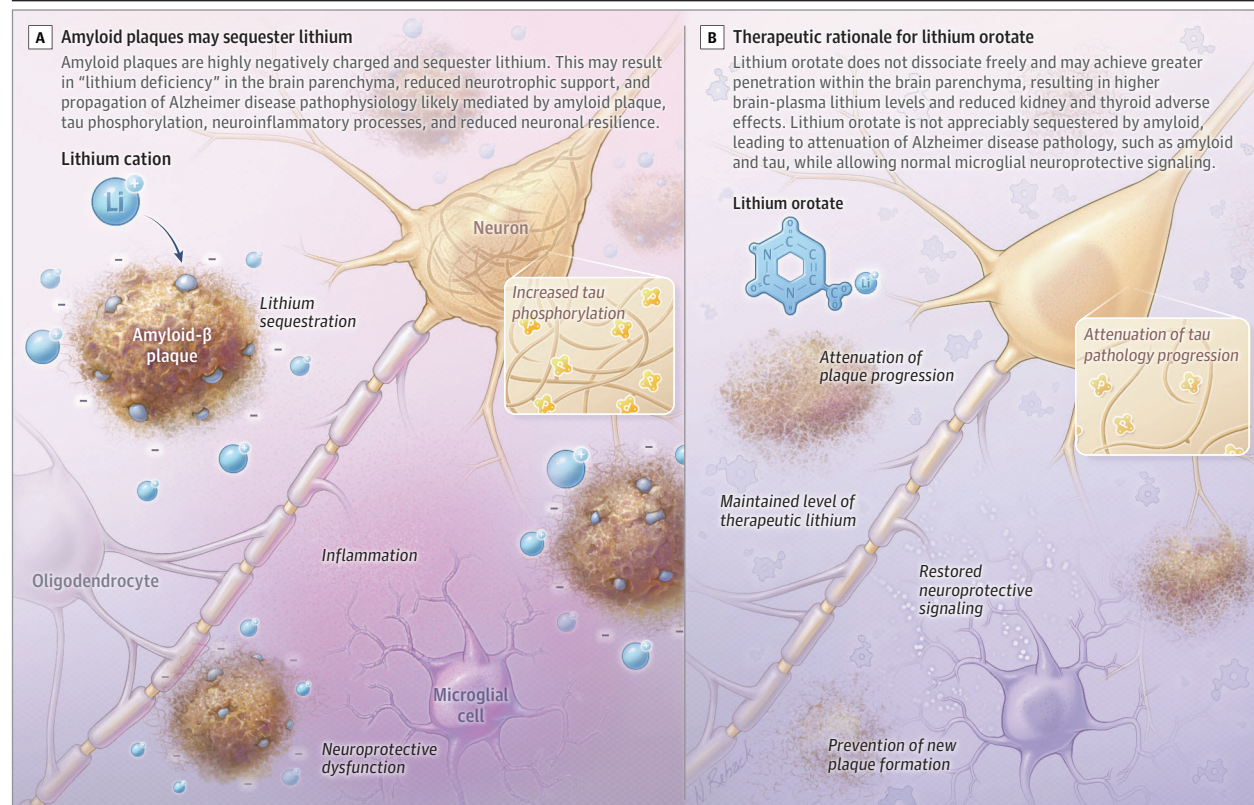
Downstream effects also include reduced tau phosphorylation and attenuation of neurodegenerative signaling. In parallel, lithium inhibits inositol monophosphatase (IMPase), depleting inositol/inositol-1,4,5-trisphosphate (IP_3) signaling and reducing endoplasmic reticulum-mediated calcium ion (Ca^{2+}) release, thereby improving intracellular calcium homeostasis. These upstream effects converge on the mitochondrion, where increased Bcl-2 stabilizes mitochondrial membranes, limits calcium overload, reduces cytochrome C release, decreases caspase activation, and enhances oxidative phosphorylation and adenosine triphosphate (ATP) production. Collectively, these coordinated actions shift intracellular signaling away from apoptosis and toward neuronal viability, metabolic stability, and synaptic plasticity, providing a mechanistic framework for lithium's disease-modifying and neuroprotective effects. Li^+ indicates lithium cations; Mg^{2+} , magnesium ions; Raf, rapidly accelerated fibrosarcoma (a serine/threonine-protein kinase); Ras, rat sarcoma virus protein.

early AD and found that lithium was depleted within vulnerable cortical and hippocampal areas. Serum and peripheral lithium levels were largely normal, suggesting a disturbance in lithium distribution.⁶⁴

Mechanistic experiments revealed an unexpected potential explanation for this paradox. Amyloid plaques were found to carry a

strong net negative charge and to act as ionic sinks that selectively bind positively charged lithium ions. Within plaque cores, lithium concentrations were considerably higher than in adjacent healthy tissue.⁶⁴ Thus, amyloid pathology may create islands of functional lithium deprivation precisely within the circuits most susceptible to degeneration (Figure 2).⁶⁴⁻⁶⁶

Figure 2. Putative Lithium Deficiency as a Contributor to Propagating Alzheimer Disease (AD) Pathophysiology and the Therapeutic Rationale for Lithium Orotate



A, In AD, extracellular aggregates of amyloid- β (A β) peptides form negatively charged plaques—one of the key hallmarks of AD. Aron and colleagues⁶⁴ reported in 2025 that these plaques can sequester lithium cations (Li⁺), effectively reducing local lithium bioavailability in the brain parenchyma (ie, creating a localized lithium deficit in critical brain areas despite normal trace lithium plasma levels). As outlined in this article and in Figure 1, lithium supports neuronal resilience through multiple neurotrophic and cytoprotective signaling pathways, including modulation of brain-derived neurotrophic factor (BDNF), B-cell lymphoma 2 (Bcl-2), glycogen synthase kinase-3 (GSK-3) inhibition, and mitochondrial stabilization. Consequently, regional brain parenchymal lithium

depletion may diminish neurotrophic support and increase neuronal vulnerability to metabolic stress, inflammation, and proteotoxic injury, thereby accelerating neurodegeneration and cognitive decline. B, To restore brain lithium levels while minimizing plaque sequestration, Aron and colleagues⁶⁴ evaluated lithium orotate, a salt with lower ionization and reduced nonspecific binding to negatively charged aggregates. In preclinical models, lithium orotate treatment attenuated amyloid plaque accumulation, potentially by enhancing microglial-mediated A β clearance, and preserved synaptic integrity and cognitive performance in mouse models of AD and normal aging. For a detailed discussion, see cited references.⁶⁴⁻⁶⁶

In mouse models, dietary lithium restriction was found to accelerate amyloid deposition, increase tau hyperphosphorylation, amplify microglial dysfunction, and worsen synaptic loss. Conversely, restoring physiological brain lithium levels attenuated these processes.⁶⁴ These findings suggest that lithium depletion may not simply accompany neurodegeneration but may facilitate disease progression by removing endogenous trophic support.

Lithium Across Neurodegenerative Disorders

While it is beyond the scope of this narrative review to cover lithium's potential utility across all neurodegenerative disorders, scientific rationale (and some data) suggest potential benefit beyond AD. The rationale derives from lithium's aforementioned neuroprotective mechanisms that intersect with pathogenic processes across multiple fundamental neurodegenerative processes. Lithium may also interact with disease-specific proteinopathies. In dementia with Lewy bodies, lithium-induced autophagy may facilitate clearance of α -synuclein aggregates,⁶⁷⁻⁶⁹ and experimental synucleinopathy models have shown reduced α -synuclein accumulation and neuro-

degeneration with lithium exposure.^{69,70} Chronic lithium also reduces tau hyperphosphorylation and neurofibrillary pathology in tauopathy models.^{71,72} In vascular cognitive impairment, lithium's effects on mitochondrial function, oxidative stress, and neuroinflammation may mitigate ischemia-related neuronal injury and post-ischemic cognitive decline.⁷³⁻⁷⁷ (Table 1).

As noted previously, AD may involve both these shared mechanisms and an additional disease-specific interaction, with lithium being depleted as amyloid- β plaques sequester lithium, reducing bioavailable lithium in plaque-adjacent tissue.⁶⁴ Taken together, lithium's effects in dementia may reflect both baseline neurotrophic support and pathology-specific interactions. While core mechanisms likely extend across multiple dementias, disease-specific factors may also influence therapeutic responsiveness.

Lithium Formulation and Dose-Dependent Effects

An additional translational question raised by recent work concerns so-called lithium subtypes and whether different lithium salts have distinct pharmacological or neurobiological effects. Lithium car-

Table 1. Mechanistic Rationale and Pathological Targets for Lithium Therapy in Major Dementia Subtypes

Dementia type	Dominant pathology	Key mechanism of lithium action	AD-specific sequestration?	Strength of rationale
AD	Amyloid β plaques, tau tangles, synaptic loss	Inhibits GSK-3 β to reduce tau; upregulates BDNF/TrkB and Bcl-2 for neuroprotection	Yes: amyloid β plaques may sequester endogenous lithium, lowering brain bioavailability	High (strongest mechanistic and clinical hypothesis)
Dementia with Lewy bodies	α -Synuclein aggregation, mitochondrial decay	Promotes autophagic clearance of α -synuclein; reduces synuclein-associated cell death	No known mechanism	Moderate (extrapolated from Parkinson disease models)
Frontotemporal dementia (FTD)	Tau or FTD pathology	Targets GSK-3 β to reduce tau hyperphosphorylation and tangle formation (in tau-positive forms)	No established mechanism	Moderate (tau positive); low (TDP-43)
Vascular dementia	Hypoperfusion, BBB dysfunction, inflammation	Mitigates ischemic injury via BDNF/TrkB and anti-apoptotic signaling; reduces neuroinflammation	No	Moderate (supported by ischemia or stroke models)
Mixed dementia	Combined AD + vascular or Lewy pathology	Pleiotropic profile: addresses tau, inflammation, and vascular injury simultaneously	Possibly (related to the AD component)	Strong (conceptually); limited clinical data

Abbreviations: AD, Alzheimer disease; BBB, blood-brain barrier; Bcl-2, B-cell lymphoma 2; BDNF, brain-derived neurotrophic factor; GSK-3 β , glycogen synthase kinase β ; TDP-43, transactive response DNA-binding protein 43 kDa; TrkB, tropomyosin receptor kinase B.

bonate remains the standard formulation used in psychiatry and where most clinical evidence has been generated. In contrast, lithium orotate has gained attention in the supplement and nootropic space and is typically marketed at substantially lower elemental lithium doses than those used in psychiatric treatment. Notably, despite being widely regarded as the gold standard treatment for BD, lithium is often not used because of its narrow therapeutic index and frequency of adverse effects at higher levels. Thus, the possibility that lithium may exert neurotrophic effects and long-term benefit at lower doses has generated considerable excitement.

Interest in lithium orotate partly derives from early hypotheses that the orotate moiety might facilitate cellular transport, potentially allowing central nervous system (CNS) effects at lower lithium exposures.⁷⁸ The suggestion that lithium orotate may cross the blood-brain barrier (rather than just lithium cations [Li⁺]) may seem counterintuitive; however, some data support this. One hypothesis is the formation of a contact ion pair (or “inner-sphere complex”) in which the lithium cation and orotate anion remain closely associated, effectively behaving as a partially covalent unit (see Lowe⁷⁹). In this state, the ions are not fully dissociated in solution. Furthermore, some evidence suggests that orotate salts may behave differently than common salts. For example, magnesium orotate lacks the laxative effects typical of other magnesium salts, suggesting reduced dissociation under physiological conditions.⁷⁹ Additionally, Aron and colleagues⁶⁴ found that lithium orotate solutions exhibited lower conductivity than other lithium salts, consistent with greater ion pairing. Together, these findings raise the possibility that lithium orotate may be transported as an intact ion pair,

potentially altering its pharmacokinetics and facilitating CNS delivery. This provides a rationale for considering lithium orotate as distinct from other lithium salts and as a candidate for targeted CNS supplementation.

Aron and colleagues⁶⁴ demonstrated that lithium orotate has lower affinity for amyloid aggregates than lithium carbonate; it also appeared to achieve greater penetration into nonplaque brain regions.^{78,80,81} In preclinical AD models, microdose lithium orotate restored brain lithium concentrations, reduced plaque burden, and improved memory performance without systemic toxicity.⁶⁴ However, human pharmacokinetic, safety, and efficacy data remain limited. Clarifying whether different lithium salts differ meaningfully in bioavailability, tissue distribution, or engagement of lithium-sensitive signaling pathways could have implications for both psychiatric treatment and emerging neuroprotective strategies (Table 2).

Conclusions and Future Directions

Across molecular, cellular, imaging, epidemiologic, and early clinical domains, lithium displays remarkable convergence across translational domains.^{5-7,12,13,42,55-58} Yet compelling biology alone is insufficient. Definitive, adequately powered randomized clinical trials are required to determine whether lithium meaningfully slows cognitive decline. While trace-level repletion is an intriguing concept, the depth of the evidence regarding the neuroprotective efficacy of approximately 0.3-mM concentrations provides a more robust

Table 2. Pharmacological Profiles and Preclinical and Clinical Evidence for Standard, Supplemental (Especially Lithium Orotate), and Low-Dose Lithium^a

Evidence	Lithium carbonate	Lithium orotate	Low-dose strategies
Preclinical evidence	Abundant data from rodent and in vitro studies showing BDNF and Bcl-2 upregulation, GSK-3 β inhibition, and neuroprotection against diverse insults. All studies at approximately 0.8mM. Some in vitro studies used LiCl as the lithium source	In Alzheimer mouse models: restoration of lithium levels in the brain, reduction of amyloid plaque burden, improvement in memory performance without systemic toxicity or cognitive adverse effects	Some rodent and in vitro studies have demonstrated Bcl-2 upregulation and neuroprotection
Clinical evidence	Extensive clinical evidence for antimanic effects, antidepressant-potential effects, long-term mood stabilization, suicide prevention, and improved outcomes overall	Preclinical Alzheimer models show reduced plaques and improved memory; limited clinical data	Independent clinical studies support the safety and rationale for low-dose strategies: Cognitive tolerability: low-dose lithium (serum 0.25-0.5 mEq/L) well tolerated in Forlenza aMCI trials Microdosing viability: 0.3 mg/d of elemental lithium with zero dropouts. Long-term outcomes: sustained exposure with improved cognitive trajectories for up to 13 y
Blood-brain barrier penetration	Passive diffusion	Active transport via URAT1	Unknown
Binding to amyloid plaques	High, but this finding needs independent replication	Low	Unknown
Effective dose (elemental lithium)	Approximately 300-900 mg/d (plasma levels approximately 0.8mM in bipolar disorder; lower levels effective as antidepressant-potentiator)	Approximately 1.5-5 mg/d; needs independent replication	Generally <0.5 mEq/L serum (eg, 0.3 mg/d); needs more robust validation
Kidney or thyroid effects risk	Moderate to high	Negligible at microdoses	No significant deterioration observed in long-term low-dose administration compared to placebo
Neuroprotective scope	Potentially quite broad given impact on BDNF, Bcl-2/GSK-3 (see Table 1)	Prominent effects in preclinical Alzheimer disease models (needs independent replication)	Unknown
Target plasma level	0.6-1.0 mEq/L	<0.1 mEq/L	0.25-0.5 mEq/L
Key advantages	Approximately 70 y of clinical efficacy data in bipolar disorder. Very well-characterized safety data over decades of administration and pharmacokinetics	Hypothesis of improved cellular transport or distribution (nonsequestration by amyloid plaques).	Greater tolerability, better cognitive trajectory, lower risk of toxicity for patients with neurodegeneration
Limitations	Narrow therapeutic index; requires serum monitoring	Lack of controlled clinical trials; incomplete safety profile	Limited understanding of optimal dosing strategies

Abbreviations: Bcl-2, B-cell lymphoma 2; BDNF, brain-derived neurotrophic factor; GSK-3, glycogen synthase kinase 3; LiCl, lithium chloride; URAT1, urate transporter.

SI conversion factor: To convert lithium from mEq/L to mmol/L, multiply by 1.

^a This consolidated format highlights the fact that patients with neurodegeneration may be vulnerable to central nervous system adverse effects, emphasizing the justification for low-dose strategies (0.25-0.5 mEq/L) that avoid cognitive clouding and the kidney or thyroid issues associated with standard bipolar doses of lithium carbonate.

foundation for clinical intervention (see sources cited here⁴⁸⁻⁵³ and the references therein).

While brain levels of elemental lithium may be the key consideration, low-dose lithium orotate may be an ideal candidate for such clinical trials. If this formulation results in a higher brain to plasma ratio, it could optimize neuroprotective delivery while reducing peripheral adverse effects. Key research priorities include (1) human dose-response studies to verify the pharmacokinetics and brain delivery of

the orotate formulation; (2) safety monitoring specifically at these sub-therapeutic levels; and (3) biomarker-guided intervention to identify patients most likely to benefit from early treatment.

If validated, lithium could represent a rare example of a low-cost, mechanistically grounded, disease-modifying therapy for neurodegeneration. In short, 75 years after transforming psychiatric care, lithium is now at the forefront of a new frontier: the fight against dementia.

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